Lecture28_MA5104

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Lecture28_ MA5104

MA 5104: Statistical Inference and Learning

Module 4: Learning
Lecture 28: Finishing the Linear Model, and Estimating Test Error

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An Example Before a Definition

A plant runs a process at two temperatures with two catalysts. Mean yield (%) in each of the four conditions:

	160°C	180°C
Catalyst A	60	78
Catalyst B	72	66

What is the effect of raising the temperature by 20°?

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An Example Before a Definition

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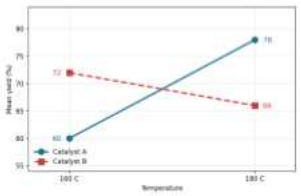
Averaging the two catalysts gives +12. That is correct arithmetic and useless advice: the true answer is +18 for catalyst A and −6 for catalyst B.

There is no such thing as "the effect of temperature" here. The question is malformed until you say *which catalyst*.

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Avg over catalyst values is not meaningful bcoz you are saying "no matter what catalyst you use"

Interaction: When the Lines Are Not Parallel



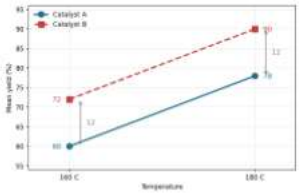
B beats A by 12 at 160° and *loses* by 12 at 180°. N

An interaction means the effect of one variable depends on the value of another. Parallel lines let you summarise each factor on its own; non-parallel lines do not, and averaging will not rescue you.

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The Additive Case: When Separate Summaries Work

Same experiment, one number changed: catalyst B at 180° now yields 90 instead of 66.



- Raising the temperature is worth +18 — for *both* catalysts.
- Catalyst B beats A by 12 — at *both* temperatures.

Two sentences, each true without conditions.

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What "Linear Model" Actually Means

Linear in the parameters β_j, not necessarily in the original inputs. Sources of columns of X:

Transformations

$X_j = \log x, \quad \sqrt{x}, \quad x^2$

Categorical variables

$X_j = \mathbf{1}(G = j)$

Basis expansions

$X_1 = x, \quad X_2 = x^2, \quad X_3 = x^3$

Interactions

$X_3 = X_1 \cdot X_2$

Every one of these is a column of X.

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Categorical Predictors: Dummy Coding

A predictor with k levels enters as $k-1$ indicator columns, with one level held out as the reference.

Three departments, reference = Maths:

$$\hat{y} = \beta_0 + \beta_1 \mathbf{1}\{\text{Physics}\} + \beta_2 \mathbf{1}\{\text{Chemistry}\} + \cdots$$

so β_1 is the Physics–Maths difference, not a Physics "effect".

- Why $k-1$ and not k: with all k columns plus an intercept, the columns sum to the intercept column. X loses rank and we are back on the previous failure slide.
- Changing the reference level changes every coefficient and changes no

- Why $k - 1$ and not k: with all k columns plus an intercept, the columns sum to the intercept column. X loses rank and we are back on the previous failure slide.
- Changing the reference level changes every coefficient and changes no fitted value whatsoever.
- Ordinal predictors: coding as $1, 2, 3$ asserts equal spacing. That is an assumption, and usually a wrong one.

Write the correct model between

Education — 10th, 12th, BSc, MSc, PhD
Occupation-type — Blue collar, White collar

income (Y) = f(education (X_1), occupation_types (X_2))

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 \cdot X_2 + \epsilon$$

The common slope model-(assuming no interactions!)

The regression model for income (Y) v.s. (education X and occupation types D_1)

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_{1i} + \varepsilon_i, \quad i = 1, \ldots, n.$$

- For Blue collar group ($D_1 = 0$):

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

- For White collar group ($D_1 = 1$):

$$Y_i = (\beta_0 + \gamma_1) + \beta_1 X_i + \varepsilon_i,$$

- Here, both groups have their own regression lines. Same slope but different intercept. It means that they are parallel!)

The interaction model

The regression model for income (Y) v.s. (education X and occupation types D_1)

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_{1i} + \delta_1 (X_i D_{1i}) + \varepsilon_i, \quad i = 1, \ldots, n.$$

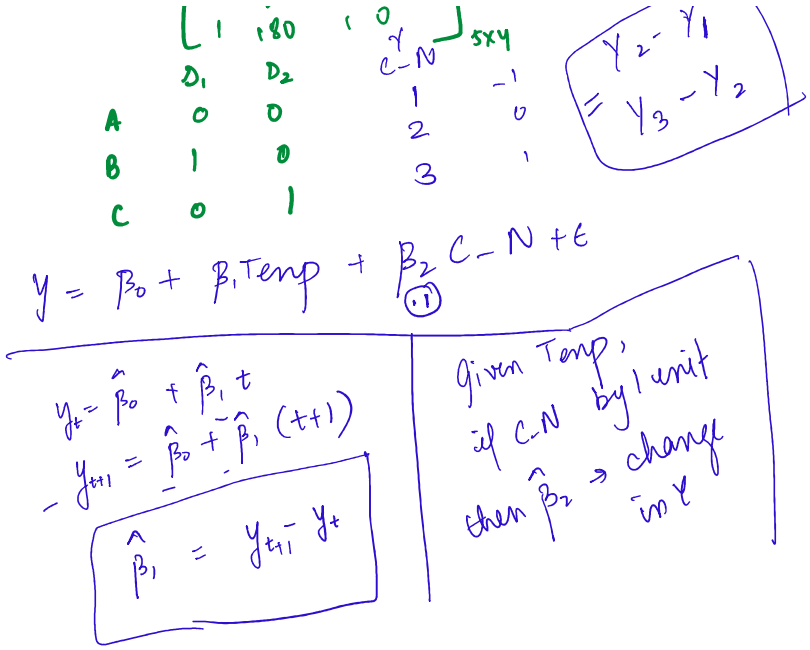
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- For White collar group ($D_1 = 1$):

$$Y_i = (\beta_0 + \gamma_1) + (\beta_1 + \delta_1) X_i + \varepsilon_i,$$

- Each group has different slope and different intercept. If there is interaction, they are NOT parallel!.
- In other words, the value of the categorical predictor also affects the association between Y(income) and X(education);



Reading the Output: A Worked Case

Starting salary (Rs. lakh p.a.) on CGPA and programme, reference = M.Sc.:

$$\widehat{\text{salary}} = 4.20 + 1.35\,\text{CGPA} + 2.10 \cdot \mathbf{1}\{\text{M.Tech}\} + 0.45\,\text{CGPA} \cdot \mathbf{1}\{\text{M.Tech}\}$$

Coef.	Reads as	Meaning
4.20	intercept	

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Coef.	Reads as	Meaning
4.20	intercept	M.Sc. student with CGPA $= 0$. Nobody. An extrapolation.
1.35	slope	

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4.20	intercept	M.Sc. student with CGPA $= 0$. Nobody. An extrapolation.
1.35	slope	per CGPA point, *for M.Sc.*
2.10	dummy	

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4.20	intercept	M.Sc. student with CGPA $= 0$. Nobody. An extrapolation.
1.35	slope	per CGPA point, *for M.Sc.*
2.10	dummy	M.Tech salary at CGPA $= 0$.
0.45	interaction	

avg. subtracting M.Sc. avg. salary

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Reading the Output: A Worked Case

fix terms

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Coef.	Reads as	Meaning
4.20	intercept	M.Sc. student with CGPA $= 0$. Nobody. An extrapolation.
1.35	slope	per CGPA point, *for M.Sc.*
2.10	dummy	M.Tech salary at CGPA $= 0$.
0.45	interaction	extra rise per CGPA point for M.Tech; their slope is 1.80.

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Validation Set

M3 : $y = \delta_0 + \delta_1 X + \delta_2 D + \epsilon$

Split once: fit on one part, score on the other.

M1 : $y = \beta_0 + \beta_1 X + \beta_2 D + \epsilon$

M2 : $y = \gamma_0 + \gamma_1 X + \gamma_2 D + \gamma_{12} XD + \epsilon$

Build M1, M2 on Train set, test on test set

80-20% ?

50-50% ?

$\| Y - \hat{Y}_{M1} \|_2$

$\| Y - \hat{Y}_{M2} \|_2$

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Validation Set

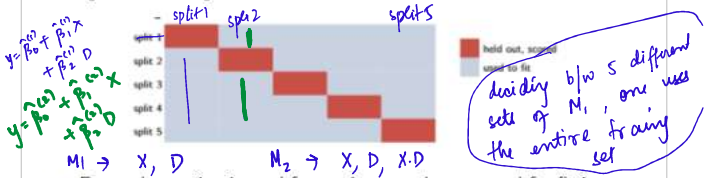
Split once: fit on one part, score on the other.

Honest, but wasteful (you fit on half the data) and

highly variable since a different split gives a different answer.

K-Fold CV

Split into K parts. Fit K times, each time holding out one part and scoring on it. Average the K errors.



$M_1 \rightarrow X, D$ $\quad M_2 \rightarrow X, D, X{:}D$

deciding b/w 5 different sets of M_1, one uses the entire training set

- Every observation is used for scoring exactly once, and for fitting $K - 1$ times.
- K has its own bias–variance trade: small K means each fit uses less data, so test error is over-estimated (bias); large K means the K training sets overlap heavily, so the K errors are correlated and the average is noisier. $K = 5$ or 10 is the usual compromise.